Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Compounds formed when the noble gases get entrapped in the cavities of crystal lattices of certain organic and inorganic compounds are known as

(A) interstitial compounds

(B) clathrates

(C) hydrates

(D) picrates

Q.2

Which of the following xenon-OXO compounds may not be obtained by hydrolysis of xenon flurides?

(A) XeO_2F_2

(B) $XeOF_4$

(C) XeO_3

(D) XeO_4

Q.3

Which of the following cannot be formed?

(A) He^{2+}

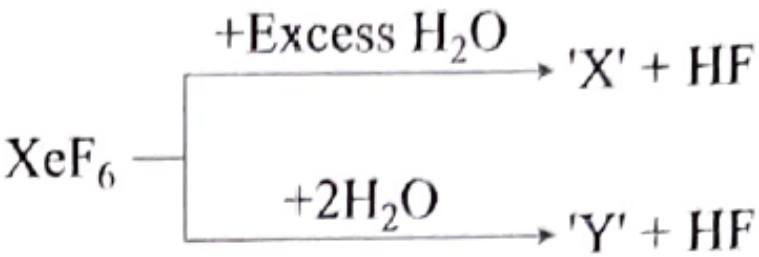
(B) $He^{\oplus}$

(C) He

(D) He_2

Q.4

Incorrect statement regarding following reactions is:



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(A) XeF_6 can undergo partial hydrolysis

(B) Both are example of non-redox reaction

(C) 'Y' is an oxyacid of xenon

(D) 'X' is explosive

Q.5

$MF + XeF_4 \longrightarrow' A' (M^+ =$ Alkali metal cation)

The state of hybridisation of the central atomin 'A' and shape of the species are:

(A) sp^3d^3, distorted octahedral

(B) sp^3d, TBP

(C) sp^3d^3, pentagonal planar

(D) No compound formed at all

Q.6

The lightest gas is

(A) N_2

(B) Ar

(C) Rn

(D) He

Q.7

Which inert gas have highest boiling point

(A) Xe

(B) Ar

(C) Kr

(D) He

Q.8

Which one of the following statements regarding helium is incorrect

(A) It is used to produce and sustain powerful superconducting magnets

(B) It is used as a cryogenic agent for carying out experiments at low temperatures

(C) It is used to fill gas balloons instead of hydrogen because it is lighter and non-inflammable

(D) It is used in gas-cooled nuclear reactors

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Q.9

Which of the following are formed by Xenon

(A) XeF_3

(B) XeF_4

(C) XeF_5

(D) None of these

Q.10

The ionization energy of $Xe \rightarrow Xe^+ + e^-$ is very close to that of

(A) $N_2 \rightarrow N_2^+ + e^-$

(B) $O_2 \rightarrow O_2^+ + e^-$

(C) $C_2 \rightarrow C_2^+ + e^-$

(D) $B_2 \rightarrow B_2^+ + e^-$

Q.11

When a mixture of Xe and F_2 in the ratio of $1:5$ is heated at $400^\circ C$ in a sealed nickel vessel, the compound formed is

(A) XeF

(B) XeF_2

(C) XeF_4

(D) XeF_6

Q.12

The number of paired electrons around Xe in XeF_2 is

(A) 3

(B) 4

(C) 5

(D) 6

Q.13

The structure of XeO_4 is

(A) square planar

(B) tetrahedral

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(C) pentagonal with one paired electrons

(D) octahedral with two paired electrons

Q.14

Which of the following statements is incorrect?

(A) The processes $\mathrm{Xe(g) \rightarrow Xe^{+}(g) + e^{-}}$ and $\mathrm{O_2(\ g) \rightarrow O_2^{+}(g) + e^{-}}$ involve more or less identical ionization energy

(B) $\mathrm{XeF_2}$ possesses an angular structure

(C) $\mathrm{XeF_2}$ possesses a trigonal bipyramidal structure with xenon and fluorine atoms occuping linear position and the three lone pairs occupying equatorial position

(D) $\mathrm{XeF_4}$ is regarded as pseudo-octahedral with fluorines occupying the corners of a square plane

Q.15

The formula of freon -12 is

(A) $\mathrm{CFCl_3}$

(B) $\mathrm{CF_2Cl_2}$

(C) $\mathrm{CF_3Cl}$

(D) $\mathrm{CF_4}$

Answers & Solutions

Q.1 Answer:

clathrates

Solution:

Clathrates (also known as cage compounds) are compounds of noble gases in which they are trapped within cavities of crystal lattices of certain organic and inorganic substances.

Q.2 Answer:

$\mathrm{XeO_4}$

Solution:

The hydrolysis of xenon fluorides proceeds as follows. $\mathrm{XeF_2 + H_2O \rightarrow Xe + \frac{1}{2}O_2 + 2HF}$ $\quad \mathrm{3XeF_4 + 6H_2O \rightarrow 2Xe + \frac{3}{2}O_2 + XeO_3 + 12HF}$

$\mathrm{XeF_6 + 3H_2O \rightarrow XeO_3 + 6HF}$ $\quad \mathrm{XeF_6 + \underset{\left(\substack{\text{small}\\\text{quality}}\right)}{H_2O} \rightarrow XeOF_4 + 2HF}$

$\mathrm{XeOF_4 + XEO_3 \rightarrow 2XeOF_2}$

Q.3 Answer:

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He_2

Solution:

He_2: $\sigma 1s^2, \sigma^* 1s^2$

Bond Order $= \frac{2-2}{2} = 0$

Not formed, does not exist

Q.4 Answer:

'Y' is an oxyacid of xenon

Solution:

(1) Correct. It can go partial hydrolysis.

(2) Both are example of non-redox reaction

(3) $Y = XeO2\ F2$**. No, it is not an oxyacid of** $Xenon$

(4) $X = XeO3$**. Yes, it is explosive**

Q.5 Answer:

sp^3d^3, pentagonal planar

Solution:

sp^3d^3**, pentagonal planar**

Q.6 Answer:

He

Solution:

The lightest gas is hydrogen (H)**. Tos atomic weight is** 1.008**. But among the mentioned gases lightest one is helium (He). Its atomic weight is** 4.003**.**

Q.7 Answer:

Xe

Solution:

	He	Ne	Ar	Kr	Xe	Rn
Boiling point of Inert gases	-269	-246	-186	-153.6	-108.1	-62

Q.8 Answer:

It is used to fill gas balloons instead of hydrogen because it is lighter and non-inflammable

Solution:

Helium is twice is heavy as hydrogen, its lifting power is 92% **of that of hydrogen. Helium has the lowest melting and boiling point of any element which makes liquid helium an ideal coolant for**

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many extremely low temperature application such as super conducting magnet and cryogenic research where temperature close to absolute zero are needed.

Q.9 Answer:

XeF_4

Solution:

XeF_4 **– This compound is formed by xenon.**

Xe compounds can contain only even no. of fluorine atoms.

Xe has a complete filled 5 p configuration. As a result when it undergoes bonding with an odd number (3 or 5) of F atoms it leaves behind one unpaired electron. This causes the molecule to become unstable.

Q.10 Answer:

$O_2 \rightarrow O_2^+ + e^-$

Solution:

$O_2 \rightarrow O_2^+ + e^-$

Q.11 Answer:

XeF_4

Solution:

XeF_4

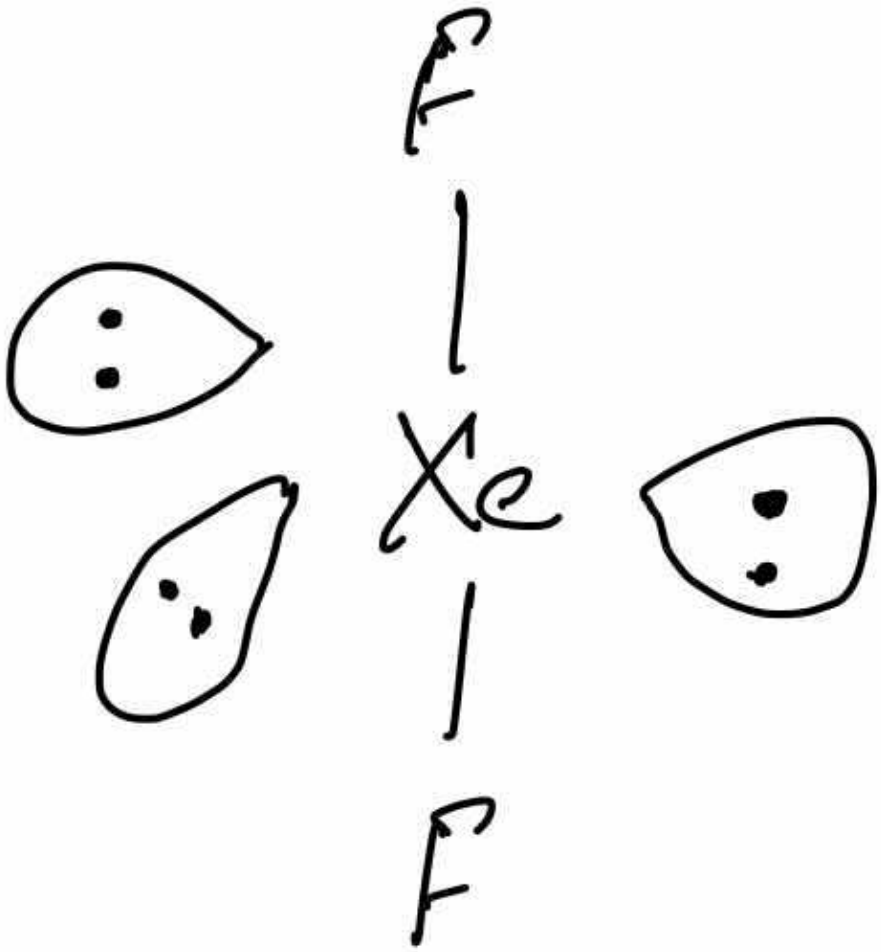
Q.12 Answer:

5

Solution:

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XeF_2 **has**



structure

No. of lone pairs $= 3$

No. of bond pairs $= 2$

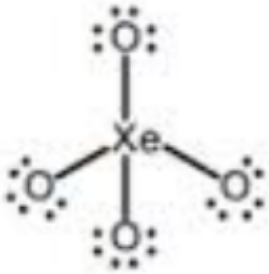
total paired electrons $= 5$

Q.13 Answer:

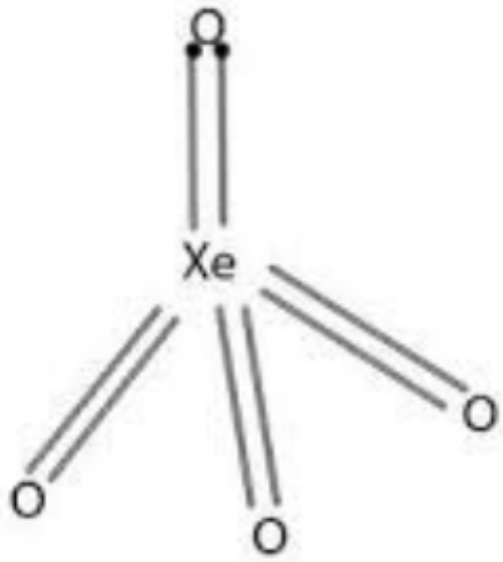
tetrahedral

Solution:

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tetrahedral; sp^3



XeO_4

Q.14 Answer:

XeF_2 possesses an angular structure

Solution:

XeF_2 **molecule has a trigonal bipyramidal structure.**

Q.15 Answer:

CF_2Cl_2

Solution:

CF_2Cl_2

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